2009 Team Answers

T-1. 135432

T-2. $2\sqrt{10}$

T-3. 32

T-4. $\frac{49}{3}$

T-5. $\left(2+\sqrt{2}, 1+\sqrt{2}\right)$

T-6. 2604

T-7. 10

T-8. $\frac{179}{250}$

T-9. $\frac{15}{2}$

T-10. 9046

2009 Team Solutions

T-1. Note that $264 = 3 \cdot 8 \cdot 11$, so we will need to address all these factors. Because the sum of the digits is 18, it follows that 3 divides N, regardless of how we order the digits of N. In order for 8 to divide N, we need N to end in $\underline{O}12, \underline{O}52, \underline{E}32$, or $\underline{E}24$, where O and E denote odd and even digits. Now write $N = \underline{U}\,\underline{V}\,\underline{W}\,\underline{X}\,\underline{Y}\,\underline{Z}$. Note that N is divisible by 11 if and only if $(U + W + Y) - (V + X + Z)$ is divisible by 11. Because the sum of the three largest digits is only 12, we must have $U + W + Y = V + X + Z = 9$.

Because Z must be even, this implies that V, X, Z are $2, 3, 4$ (in some order). This means $Y \neq 2$, and so we must have $Z \neq 4 \Rightarrow Z = 2$. Of the three remaining possibilities, $\underline{E}32$ gives the smallest solution, **135432**.

T-2. Set $\text{m}\angle ABC = x$ and $\text{m}\angle TBQ = y$. Then $x + y = 180^\circ$ and so $\cos x + \cos y = 0$. Applying the Law of Cosines to triangles ABC and TBQ gives $AC^2 = AB^2 + BC^2 - 2AB \cdot BC \cos x$ and $QT^2 = BT^2 + BQ^2 - 2BT \cdot BQ \cos y$, which, after substituting values, become $8^2 = 4^2 + 6^2 - 48 \cos x$ and $QT^2 = 4^2 + 6^2 - 48 \cos y$.

Adding the last two equations yields $QT^2 + 8^2 = 2(4^2 + 6^2)$ or $QT = \mathbf{2\sqrt{10}}$.

Remark: This problem is closely related to the fact that in a parallelogram, the sum of the squares of the lengths of its diagonals is the equal to the sum of the squares of the lengths of its sides.

T-3. We say that two numbers are *neighbors* if they occupy adjacent squares, and that a is a *friend* of b if $0 < |a - b| \leq 2$. Using this vocabulary, the problem's condition is that every pair of neighbors must be friends of each other. Each of the numbers 1 and 8 has two friends, and each number has at most four friends.

If there is no number written in the center square, then we must have one of the cycles in the figures below. For each cycle, there are 8 rotations. Thus there are 16 possible configurations with no number written in the center square.

2	1	3
4	—	5
6	8	7

3	1	2
5	—	4
7	8	6

Now assume that the center square contains the number n. Because n has at least three neighbors, $n \neq 1$ and $n \neq 8$. First we show that 1 must be in a corner. If 1 is a neighbor of n, then one of the corners neighboring 1 must be empty, because 1 has only two friends (2 and 3). If c is in the other corner neighboring 1, then $\{n, c\} = \{2, 3\}$. But then n must have three

more friends (n_1, n_2, n_3) other than 1 and c, for a total of five friends, which is impossible, as illustrated below. Therefore 1 must be in a corner.

—	1	c
n_1	n	n_2
	n_3	

Now we show that 1 can only have one neighbor, i.e., one of the squares adjacent to 1 is empty. If 1 has two neighbors, then we have, up to a reflection and a rotation, the configuration shown below. Because 2 has only one more friend, the corner next to 2 is empty and $n = 4$. Consequently, $m_1 = 5$ (refer to the figure below). Then 4 has one friend (namely 6) left with two neighbors m_2 and m_3, which is impossible. Thus 1 must have exactly one neighbor. An analogous argument shows that 8 must also be at a corner with exactly one neighbor.

1	2	—
3	n	m_3
m_1	m_2	

Therefore, 8 and 1 must be in non-opposite corners, with the blank square between them. Thus, up to reflections and rotations, the only possible configuration is the one shown at left below.

1	—	8
m		

1	—	8
2/3	4/5	6/7
3/2	5/4	7/6

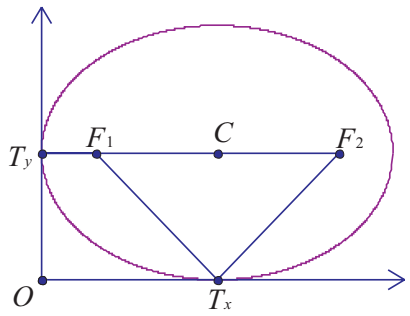
There are two possible values for m, namely 2 and 3. For each of the cases $m = 2$ and $m = 3$, the rest of the configuration is uniquely determined, as illustrated in the figure above right. We summarize our process: there are four corner positions for 1; two (non-opposite) corner positions for 8 (after 1 is placed); and two choices for the number in the square neighboring 1 but not neighboring 8. This leads to $4 \cdot 2 \cdot 2 = 16$ distinct configurations with a number written in the center square.

Therefore, there are 16 configurations in which the center square is blank and 16 configurations with a number in the center square, for a total of **32** distinct configurations.

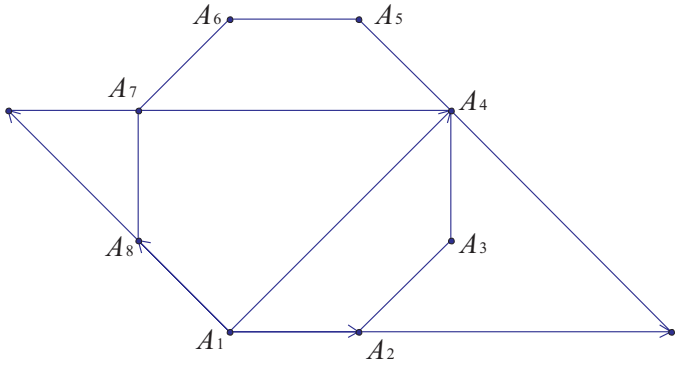
T-4. See the diagram below. The center of the ellipse is $C = \left(\frac{d+3}{2}, 7\right)$. The major axis of the ellipse is the line $y = 7$, and the minor axis is the line $x = \frac{d+3}{2}$. The ellipse is tangent to the coordinate axes at $T_x = \left(\frac{d+3}{2}, 0\right)$ and $T_y = (0, 7)$. Let $F_1 = (3, 7)$ and $F_2 = (d, 7)$. Using the locus definition of an ellipse, we have $F_1T_x + F_2T_x = F_1T_y + F_2T_y$; that is,

$$2\sqrt{\left(\frac{d-3}{2}\right)^2 + 7^2} = d + 3 \quad \text{or} \quad \sqrt{(d-3)^2 + 14^2} = d + 3.$$

Squaring both sides of the last equation gives $d^2 - 6d + 205 = d^2 + 6d + 9$ or $196 = 12d$, so $d = \mathbf{\frac{49}{3}}$.



T-5. We can scale the octagon so that $A_1A_2 = \sqrt{2}$. Because the exterior angle of the octagon is $45°$, we can place the octagon in the coordinate plane with A_1 being the origin, $A_2 = (\sqrt{2}, 0)$, and $A_8 = (1, 1)$.



Then $A_3 = (1 + \sqrt{2}, 1)$ and $A_4 = (1 + \sqrt{2}, 1 + \sqrt{2})$. It follows that $\mathbf{u} = \langle\sqrt{2}, 0\rangle$, $\mathbf{v} = \langle-1, 1\rangle$, and

$$\overrightarrow{A_1A_4} = \left\langle 1 + \sqrt{2}, 1 + \sqrt{2} \right\rangle = a\left\langle \sqrt{2}, 0 \right\rangle + b\langle -1, 1\rangle = \left\langle a\sqrt{2} - b, b \right\rangle.$$

Thus $b = \sqrt{2} + 1$ and $a\sqrt{2} - b = \sqrt{2} + 1$, or $a = 2 + \sqrt{2}$, so $(a, b) = \mathbf{(2 + \sqrt{2}, \sqrt{2} + 1)}$.

Alternate Solution: Extend $\overline{A_1A_2}$ and $\overline{A_5A_4}$ to meet at point Q; let P be the intersection of $\overline{A_1Q}$ and $\overleftrightarrow{A_6A_3}$. Then $A_1A_2 = \|\mathbf{u}\|$, $A_2P = \|\mathbf{u}\|\sqrt{2}$, and $PQ = \|\mathbf{u}\|$, so $A_1Q = \left(2 + \sqrt{2}\right)\|\mathbf{u}\|$.

Because A_1QA_4 is a $45°-45°-90°$ right triangle, $A_4Q = \dfrac{A_1Q}{\sqrt{2}} = \left(\sqrt{2}+1\right)\|\mathbf{u}\|$. Thus $\overrightarrow{A_1A_4} = \overrightarrow{A_1Q} + \overrightarrow{QA_4}$, and because $\|\mathbf{u}\| = \|\mathbf{v}\|$, we have $(a,b) = \mathbf{(2+\sqrt{2},\sqrt{2}+1)}$.

T-6. Suppose that $n = 2^k p_1^{a_1} \cdots p_r^{a_r}$, where the p_i are distinct odd primes, k is a nonnegative integer, and $a_1, \ldots, a_r$ are positive integers. Then the sum of the odd positive divisors of n is equal to

$$\prod_{i=1}^{r}(1 + p_i + \cdots + p_i^{a_i}) = \prod_{i=1}^{r}\frac{p_i^{a_i+1}-1}{p_i-1} = 1024 = 2^{10}.$$

Note that $1 + p_i + \cdots + p_i^{a_i}$ is the sum of $a_i + 1$ odd numbers. Because the product of those sums is a power of two, each sum must be even (in fact, a power of 2). Thus, each a_i must be odd.

Because $1 + 11 + 11^2 + 11^3 > 1024$, if $p_i \geq 11$, then $a_i = 1$ and $1 + p_i$ must be a power of 2 that is no greater than 1024. The possible values of p_i, with $p_i \geq 11$, are 31 and 127 (as 5 divides 255, 7 divides 511, and 3 divides 1023).

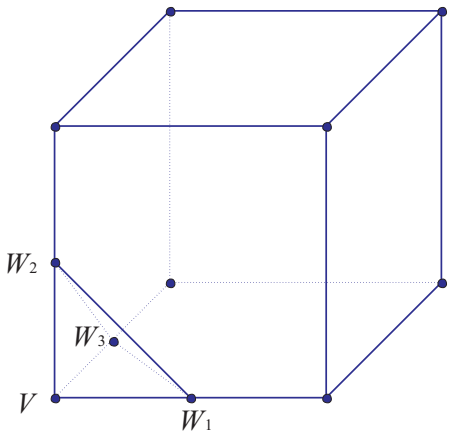
If $p_1 < 11$, then p_i can be 3, 5, 7. It is routine to check that $a_i = 1$ and $p_i = 3$ or 7.

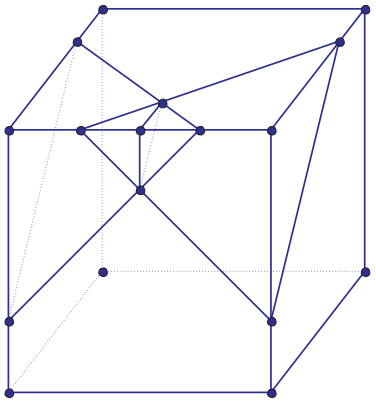
Thus $a_i = 1$ for all i, and the possible values of p_i are $3, 7, 31, 127$. The only combinations of these primes that yield 1024 are $(1+3)\cdot(1+7)\cdot(1+31)$ (with $n = 2^k \cdot 3 \cdot 7 \cdot 31 = 651 \cdot 2^k$) and $(1+7)\cdot(1+127)$ (with $n = 7 \cdot 127 = 889 \cdot 2^k$). Thus $n = 651 \cdot 2^2 = \mathbf{2604}$ is the unique value of n satisfying the conditions of the problem.

T-7. Write $x = 11 + c$ and $y = 5 + d$. Then $AR^2 = c^2 + d^2 = \frac{1}{2} \cdot 650 = 325$. Note that $325 = 18^2 + 1^2 = 17^2 + 6^2 = 15^2 + 10^2$. Temporarily restricting ourselves to the case where c and d are both positive, there are three classes of solutions: $\{c,d\} = \{18,1\}$, $\{c,d\} = \{17,6\}$, or $\{c,d\} = \{15,10\}$. In fact, c and d can be negative, so long as those values do not cause x or y to be negative. So there are **10** solutions:

(c,d)	(x,y)
$(18,1)$	$(29,6)$
$(18,-1)$	$(29,4)$
$(1,18)$	$(12,23)$
$(-1,18)$	$(10,23)$
$(17,6)$	$(28,11)$
$(6,17)$	$(17,22)$
$(-6,17)$	$(5,22)$
$(15,10)$	$(26,15)$
$(10,15)$	$(21,20)$
$(-10,15)$	$(1,20)$

T-8. For a fixed vertex V on the cube, the locus of points on or inside the cube that are at most $\frac{3}{5}$ away from V form a *corner* at V (that is, the right pyramid $VW_1W_2W_3$ in the figure shown at left below, with equilateral triangular base $W_1W_2W_3$ and three isosceles right triangular lateral faces $VW_1W_2, VW_2W_3, VW_3W_1$). Thus $\mathcal{R}$ is formed by removing eight such congruent corners from the cube. However, each two neighboring corners share a common region along their shared edge. This common region is the union of two smaller right pyramids, each similar to the original corners. (See the figure shown at right below.)





We compute the volume of $\mathcal{R}$ as

$$1 - 8 \cdot \frac{1}{6}\left(\frac{3}{5}\right)^3 + 12 \cdot 2 \cdot \frac{1}{6}\left(\frac{1}{10}\right)^3 = \mathbf{\frac{179}{250}}.$$

T-9. Each cubic expression can be depressed—that is, the quadratic term can be eliminated—by substituting as follows. Because $(a-p)^3 = a^3 - 3a^2p + 3ap^2 - p^3$, setting $p = -\frac{(-15)}{3} = 5$ and substituting $c + p = a$ transforms the expression $a^3 - 15a^2 + 20a - 50$ into the equivalent expression $(c+5)^3 - 15(c+5)^2 + 20(c+5) - 50$, which simplifies to $c^3 - 55c - 200$. Similarly, the substitution $d = b - \frac{5}{2}$ yields the equation $d^3 - 55d = -200$. [This procedure, which is analogous to completing the square, is an essential step in the algebraic solution to the general cubic equation.]

Consider the function $f(x) = x^3 - 55x$. It has three zeros, namely, 0 and $\pm\sqrt{55}$. Therefore, it has a relative maximum and a relative minimum in the interval $[-\sqrt{55}, \sqrt{55}]$. Note that for $0 \le x \le 5.5$, $|f(x)| < |x^3| < 5.5^3 = 166.375$, and for $5.5 < x \le \sqrt{55} < 8$, we have

$$|f(x)| = |x^3 - 55x| < x|x^2 - 55| < 8(55 - 5.5^2) = 198.$$

Because $f(x)$ is an odd function of x (its graph is symmetric about the origin), we conclude that for $-\sqrt{55} \le x \le \sqrt{55}$, $|f(x)| < 198$. Therefore, for constant m with $|m| > 198$, there is a unique real number x_0 such that $f(x_0) = m$.

In particular, since $200 > 198$, the values of c and d are uniquely determined. Because $f(x)$ is odd, we conclude that $c = -d$, or $a + b = \mathbf{\frac{15}{2}}$.

Alternate Solution: Set $a = x - b$ and substitute into the first equation. We get

$$\begin{aligned}(x-b)^3 - 15(x-b)^2 + 20(x-b) - 50 &= 0\\ -b^3 + b^2(3x-15) + b(-3x^2+30x-20) + (x^3 - 15x^2 + 20x - 50) &= 0\\ 8b^3 + b^2(-24x+120) + b(24x^2 - 240x + 160) - 8(x^3 - 15x^2 + 20x - 50) &= 0.\end{aligned}$$

If we equate coefficients, we see that

$$\begin{aligned}-24x + 120 &= -60\\ 24x^2 - 240x + 160 &= -290\\ -8(x^3 - 15x^2 + 20x - 50) &= 2575\end{aligned}$$

are all satisfied by $x = \frac{15}{2}$. This means that any real solution b to the second equation yields a real solution of $\frac{15}{2} - b$ to the first equation. We can follow the reasoning of the previous solution to establish the existence of exactly one real solution to the second cubic equation. Thus a and b are unique, and their sum is $(\frac{15}{2} - b) + b = \mathbf{\frac{15}{2}}$.

T-10. If $s(10x) = a$, then the values of s over $\{10x{+}0, 10x{+}1, \ldots, 10x{+}9\}$ are a, $a{+}2$, $a{+}4, \ldots, a{+}18$. Furthermore, if x is not a multiple of 10, then $s(10(x+1)) = a + 11$. This indicates that the values of s "interweave" somewhat from one group of 10 to the next: the sets alternate between even and odd. Because the s-values for starting blocks of ten differ by 11, consecutive blocks of the same parity differ by 22, so the values of s do not overlap. That is, s takes on 100 distinct values over any range of the form $\{100y + 0, 100y + 1, \ldots, 100y + 99\}$.

First determine how many values are repeated between consecutive hundreds. Let y be an integer that is not a multiple of 10. Then the largest value for $s(100y + k)$ $(0 \le k \le 99)$ is $100y + (s(y) - y) + 99 + s(99) = 100y + s(y) - y + 117$, whereas the smallest value in the next group of 100 is for

$$\begin{aligned}s(100(y+1)) &= 100(y+1) + (s(y+1) - (y+1)) = 100y + (s(y)+2) - (y+1) + 100\\ &= 100y + s(y) - y + 101.\end{aligned}$$

This result implies that the values for $s(100y + 91)$ through $s(100y + 99)$ match the values of $s(100y + 100)$ through $s(100y + 108)$. So there are 9 repeated values.
Now determine how many values are repeated between consecutive thousands. Let z be a digit, and consider $s(1000z + 999)$ versus $s(1000(z + 1))$. The first value equals

$$1000z + (s(z) - z) + 999 + s(999) = 1000z + z + 1026 = 1001z + 1026.$$

The latter value equals $1000(z+1) + (s(z+1) - (z+1)) = 1001(z+1) = 1001z + 1001$. These values differ by an odd number. We have overlap between the $982, 983, \ldots, 989$ terms and the $000, 001, \ldots, 007$ terms. We also have overlap between the $992, 993, \ldots, 999$ terms and the $010, 011, \ldots, 017$ terms, for a total of 16 repeated values in all.

There are 90 instances in which we have 9 repeated terms, and 9 instances in which we have 16 repeated terms, so there are a total of $10000 - 90 \cdot 9 - 9 \cdot 16 = \mathbf{9046}$ unique values.